

Supplementary Material

S1: Statistical Analysis and Reporting of Experimental Results

In the AVEC2014 dataset, we adopted the same fixed-length temporal normalization strategy as that used in the EIVE dataset. Before hypergraph construction and STHC feature extraction, each complete FAU sequence was linearly resampled along the temporal dimension and uniformly mapped to 60 time points. This process preserves the overall dynamic patterns of the complete sequence without direct truncation or zero-padding, thereby avoiding additional biases introduced by varying sequence lengths or padding boundaries. It is worth noting that the average sequence length of the original FAU recordings in AVEC 2014 is approximately 54 seconds. This setting not only maintains consistency with the sequence length used in the EIVE dataset but also closely matches the intrinsic temporal duration of AVEC 2014. Therefore, it minimizes potential temporal structural distortions caused by excessive compression or expansion of the original sequences.

After temporal normalization, all samples share an identical temporal dimension. The hyperedge statistical features, including the mean, standard deviation, minimum, and maximum values, are calculated based on the same number of temporal sampling points. Therefore, variations in the original video duration do not directly affect the dimensionality of the hyperedge features or the input length of STHC, thereby improving the comparability of temporal representations across different subjects. We further investigated the impact of different temporal sequence lengths on model performance by setting the normalized FAU sequences to 30, 60, and 90 temporal points, respectively. The results are presented in [Tab.1](#). The experimental results demonstrate that DEP-FAUs exhibits good robustness to the temporal length of normalized sequences. Compared with 30 temporal points, increasing the sequence length to 60 temporal points improved the ACC and F1 scores of the Northwind task from 0.700 and 0.700 to 0.720 and 0.730, respectively. Similarly, for the Freeform task, ACC and F1 increased from 0.600 and 0.610 to 0.620 and 0.630, respectively. When the temporal length was further increased to 90 points, the overall performance remained largely consistent with that achieved using 60 points, with only a slight decrease in the F1 score of the Northwind task to 0.720. These results indicate that the model performance remains stable across different temporal sequence lengths.

Tab.1. Effect of different temporal lengths on model performance in cross-dataset evaluation.

AVEC 2014/AUS	Northwind		Freeform	
Temporal length	ACC	F1	ACC	F1
30	<u>0.700</u>	0.700	<u>0.600</u>	<u>0.610</u>
60	0.720	0.730	0.620	0.630
90	0.720	<u>0.720</u>	0.620	0.630

S2: Analysis of Model Robustness, Statistical Significance, and Clinical Severity Association

In this section, we further investigate the following aspects to validate the reliability and clinical applicability of DEP-FAUs:

- a.** How stable are the reported results across different random splits?
- b.** What are the confidence intervals or statistical variations of the reported metrics?
- c.** Is the observed performance improvement statistically significant?
- d.** How does the model performance vary according to depression severity?

Regarding concern **(a)**, namely that different random data splits may lead to fluctuations in model performance, we further evaluated the stability of DEP-FAUs using repeated five-fold cross-validation with 10 different random seeds (10×5-fold cross-validation). Specifically, in each repeated experiment, DEP-FAUs and all comparison methods were evaluated using exactly the same data partitions to ensure a fair comparison. The mean and standard deviation of model performance were then calculated based on the results from the 10 repeated experiments. As shown in **Tab. 3**, DEP-FAUs maintained stable performance across different random splits. For example, under the Positive, Neutral, and Negative emotional conditions, DEP-FAUs achieved ACC values of 0.756 ± 0.021 , 0.693 ± 0.027 , and 0.733 ± 0.017 , respectively, and F1 scores of 0.763 ± 0.014 , 0.696 ± 0.021 , and 0.729 ± 0.026 , respectively. Compared with the original five-fold cross-validation results, only minor performance variations were observed across different random partitions, indicating that the performance advantages of DEP-FAUs are not dependent on any specific data split.

Regarding concern **(b)**, namely the uncertainty of model performance, we further reported the standard deviation and 95% confidence intervals (CIs) of DEP-FAUs based on the original five-fold cross-validation results. As shown in **Tab.2**, DEP-FAUs achieved ACC values of 0.756 ± 0.015 , 0.700 ± 0.017 , and 0.731 ± 0.017 under the Positive, Neutral, and Negative conditions, respectively. The corresponding 95% CIs were [0.7374, 0.7746], [0.6787, 0.7213], and [0.7101, 0.7525], respectively. In addition, the corresponding confidence intervals for other evaluation metrics, including F1 score, Recall, and Precision, were also reported. These results demonstrate that the performance variations of DEP-FAUs across different testing folds remained within a reasonable range, further validating the reliability of the model predictions.

Regarding concern **(c)**, to determine whether the performance improvements of DEP-FAUs over existing methods were statistically significant, we conducted paired statistical analyses based on the results of 10 repeated five-fold cross-validation experiments. As shown in **Tab.3**, under each emotional condition, the best-performing representative baseline model was selected (ModernTCN for the Positive condition, and CrossFormer for the Neutral and Negative conditions). The performance differences between DEP-FAUs and the corresponding baseline models were evaluated using the paired

Wilcoxon signed-rank test under identical random data splits, with the Holm method applied for multiple comparison correction. The results showed that, under the Positive emotional stimulus condition, DEP-FAUs achieved statistically significant improvements over ModernTCN in both ACC and F1 metrics (Holm-adjusted $p < 0.05$). Similarly, under the Negative emotional stimulus condition, DEP-FAUs demonstrated significant performance advantages compared with CrossFormer. For the Neutral emotional stimulus condition, the performance improvement did not reach statistical significance after multiple comparison correction, likely because facial behavioral differences between patients and controls are relatively weaker under neutral stimuli, resulting in a limited improvement margin. Nevertheless, DEP-FAUs still maintained a stable performance advantage. These results indicate that the improvements achieved by DEP-FAUs are not attributable to a single random data split but instead demonstrate robust statistical stability across different data partitions.

Regarding concern (d), to further investigate the relationship between DEP-FAUs and clinical depression severity, we conducted a subgroup analysis of 80 patients with depression based on their HAMD scores. According to clinical severity, the patients were categorized into three subgroups: subthreshold or mild depression (HAMD score: 8–20), moderate depression (HAMD score: 20–35), and severe depression (HAMD score >35). Since DEP-FAUs models FAU dynamic patterns separately under different emotional stimuli, the severity analysis was performed at the subject level. Specifically, for each patient with depression, the FAU sequences under the Positive, Neutral, and Negative emotional stimulus conditions were individually input into the model to obtain the corresponding depression prediction probabilities. The prediction probabilities from the three emotional conditions were then averaged for each subject to generate the final depression risk score, which was subsequently used for severity stratification based on HAMD scores. The results are presented in Tab.4. As the HAMD score increased, both the model discrimination ability and prediction confidence showed a gradual increasing trend. The sensitivities for patients with mild, moderate, and severe depression were 0.667, 0.778, and 0.824, respectively, with corresponding mean prediction probabilities of 0.624 ± 0.176 , 0.731 ± 0.151 , and 0.816 ± 0.128 , respectively. Furthermore, a significant positive correlation was observed between HAMD scores and model prediction probabilities (Spearman $r = 0.42$, $p < 0.001$), indicating that DEP-FAUs not only enables depression state classification but also captures facial behavioral patterns associated with clinical symptom severity.

Tab.2. Comparative experimental performance of different models on the EIVE dataset under various emotional stimuli. The symbol ‘ ’ denotes the second-best performing model.

EIVE/AUS	Positive				Neutral				Negative			
Model	Acc	F1	Rec	Pre	Acc	F1	Rec	Pre	Acc	F1	Rec	Pre
Original FAUs data												

Informer	0.731± 0.041	0.722±0.044	0.700±0.048	0.770 ±0.045	0.643±0.047	0.605±0.048	0.575±0.042	0.719 ±0.050	0.687±0.054	0.656±0.057	0.662±0.041	0.695±0.052
PatchTST	0.662± 0.058	0.696±0.051	0.787±0.068	0.648±0.060	0.606±0.035	0.544±0.054	0.537±0.021	0.648±0.046	0.637±0.055	0.634±0.051	<u>0.737</u> ±0.047	0.694±0.043
CrossFormer	0.706±0.040	0.659± 0.067	0.625±0.052	0.730±0.044	<u>0.693</u> ±0.034	<u>0.698</u> ±0.045	0.725 ±0.051	0.682±0.045	<u>0.718</u> ±0.051	<u>0.719</u> ±0.055	0.725±0.094	0.722±0.059
Mamba	0.693±0.049	0.680±0.053	0.662±0.046	0.705±0.066	0.656±0.046	0.608±0.059	0.575±0.055	<u>0.710</u> ±0.061	0.675±0.055	0.644±0.057	0.650±0.047	0.697±0.061
ModernTCN	<u>0.737</u> ±0.052	0.741±0.041	0.750±0.068	0.738±0.059	0.668±0.035	0.676±0.047	<u>0.712</u> ±0.055	0.675±0.024	0.700±0.060	0.710±0.058	0.762 ±0.046	0.681±0.057
TimeKAN	0.637±0.051	0.610±0.047	0.600±0.051	0.656±0.050	0.575±0.040	0.607±0.061	0.687±0.069	0.573±0.068	0.612±0.038	0.600±0.054	0.625±0.063	0.620±0.032
DLinear	0.675±0.047	0.654±0.037	0.612±0.052	0.712±0.050	0.643±0.017	0.635±0.028	0.625±0.056	0.655±0.066	0.643±0.068	0.622±0.055	0.587±0.061	0.661±0.057
TimesNet	0.718±0.035	0.713±0.043	0.700±0.058	<u>0.751</u> ±0.067	0.656±0.026	0.666±0.048	0.700±0.018	0.652±0.040	0.631±0.047	0.543±0.042	0.537±0.041	0.737±0.052
Hypergraph data												
STGCN	0.706±0.029	0.737±0.037	<u>0.825</u> ±0.041	0.667±0.039	0.649±0.037	0.626±0.043	0.587±0.048	0.672±0.051	0.681±0.038	0.634±0.044	0.575±0.041	0.751 ±0.039
HyperGCN	0.712±0.031	0.747±0.039	0.850 ±0.041	0.668±0.045	0.643±0.042	0.617±0.038	0.575±0.049	0.667±0.041	0.668±0.051	0.662±0.053	0.650±0.047	0.675±0.045
Mgrr-net	0.700±0.028	0.727±0.033	0.800±0.047	0.666±0.041	0.631±0.042	0.619±0.051	0.600±0.051	0.642±0.048	0.662±0.045	0.653±0.051	0.637±0.055	0.670±0.047
Dhyper	0.718±0.039	0.742±0.047	0.815±0.042	0.684±0.046	0.662±0.029	0.651±0.037	0.637±0.039	0.671±0.035	0.693±0.053	0.701±0.049	0.725±0.044	0.682±0.039
GraphEx	0.725±0.041	<u>0.749</u> ±0.037	<u>0.825</u> ±0.033	0.687±0.046	0.668±0.048	0.656±0.039	0.637±0.052	0.681±0.041	0.674±0.039	0.673±0.047	0.675±0.046	0.674±0.036
DEP-FAUs	0.756 ±0.015	0.761 ±0.018	0.787±0.025	0.746±0.022	0.700 ±0.017	0.704 ±0.020	0.725 ±0.026	0.701±0.058	0.731 ±0.017	0.726 ±0.056	<u>0.737</u> ±0.032	<u>0.744</u> ±0.088
DEP-FAUs (CI)	[0.7374, 0.7746]	[0.7387, 0.7833]	[0.7560, 0.8180]	[0.7187, 0.7733]	[0.6787, 0.7213]	[0.6780, 0.7300]	[0.6924, 0.7576]	[0.6154, 0.7603]	[0.7101, 0.7525]	[0.6565, 0.7963]	[0.6969, 0.7781]	[0.6087, 0.8281]

Tab.3. Model stability and statistical significance analysis based on **10 repeated five-fold cross-validation**.

Emotion	Metric	DEP-FAUs	95% CI	Baseline	95% CI	Wilcoxon pp	Holm pp
Positive	ACC	0.756 ± 0.021	[0.7410, 0.7710]	0.734 ± 0.044 ModernTCN	[0.7025, 0.7655]	0.0039	0.0195
	F1	0.763 ± 0.014	[0.7530, 0.7730]	0.735 ± 0.035 ModernTCN	[0.7100, 0.7600]	0.0020	0.0117
Neutral	ACC	0.693 ± 0.027	[0.6737, 0.7123]	0.690 ± 0.020 CrossFormer	[0.6757, 0.7043]	0.0742	0.1484
	F1	0.696 ± 0.021	[0.6810, 0.7110]	0.687 ± 0.029 CrossFormer	[0.6663, 0.7077]	0.0469	0.0938
Negative	ACC	0.733 ± 0.017	[0.7208, 0.7452]	0.731 ± 0.031 CrossFormer	[0.7088, 0.7532]	0.0195	0.0390
	F1	0.729 ± 0.026	[0.7104, 0.7476]	0.724 ± 0.029 CrossFormer	[0.7033, 0.7447]	0.0098	0.0293

Tab.4. Model performance analysis across different depression severity levels stratified by HAMD scores.

HAMD	n	Sensitivity	Mean prediction probability
8-20	18	0.667	0.624±0.176
20-35	45	0.778	0.731±0.151
>35	17	0.824	0.816±0.128